MA378 Chapter 3: Numerical Integration

§3.3 Precision and Composition

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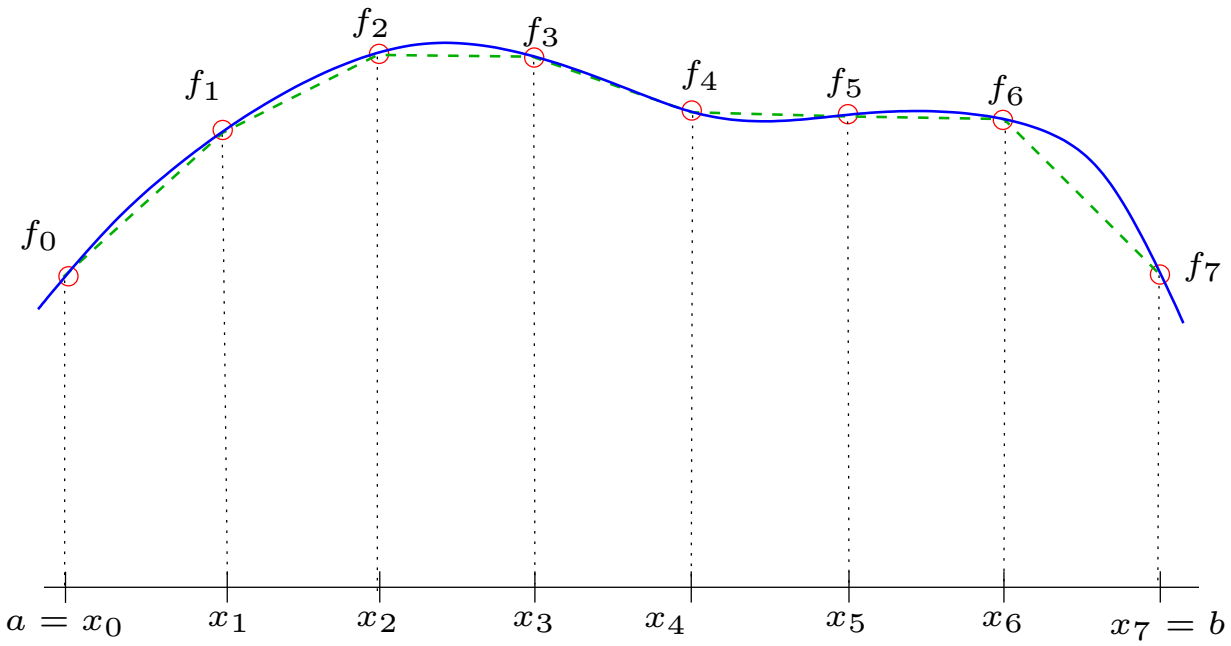


3.0 What, precisely, we'll discuss

1 Simpson's, again

- Details
- Errors

2 Precision

3 Composite Rules

4 Exercises

3.1 Simpson's, again

In the final example of Section 3.2 ("Simpson's Rule") we noticed that Simpson's Rule yields exactly the right answer (i.e., the error is zero) when applied to approximating $\int_0^1 x^3 dx$.

In fact, Simpson's Rule is exact for **any** polynomial of degree 3 or less. We'll now explain why, restricting our focus to the interval $[-1, 1]$. (See below for the general case).

1. Suppose that $Q_2(f)$ approximates $\int_{-1}^1 f(x)dx$.
2. We know that, if p_2 is any quadratic polynomial, then $Q_2(p_2) = \int_{-1}^1 p_2(x)dx$.
3. Write p_3 as

$$p_3(x) = c_3x^3 + c_2x^2 + c_1x + c_0 = c_3x^3 + p_2(x),$$

for some quadratic $p_2(x)$.

3.1 Simpson's, again

4. Note that $\displaystyle\int_{-1}^{1} p_3(x)dx = c_3 \int_{-1}^{1} x^3 dx + \int_{-1}^{1} p_2(x)dx.$
5. However, $\int_{-1}^{1} x^3 dx = 0.$
6. Therefore, $\displaystyle\int_{-1}^{1} p_3(x)dx = \int_{-1}^{1} p_2(x)dx.$
7. Similarly, $Q_2(p_3)dx = c_3 Q_2(x^3) + Q_2(p_2).$
8. However, $Q_2(x^3) = \frac{2}{6}\big((-1)^3 + 4(0^3) + 1^3\big) = 0.$
9. Therefore $Q_2(p_3)dx = Q_2(p_2) = \displaystyle\int_{-1}^{1} p_2(x)dx.$
10. So, we have that

$$\int_{-1}^{1} p_3(x)dx = \int_{-1}^{1} p_2(x)dx = Q_2(p_3).$$

Conclude: $Q_2(\cdot)$ gives exactly the right answer when applied to estimating $\int_{-1}^{1} p_3(x)dx$ for any cubic polynomial, p_3.

3.1 Simpson's, again

Details

Here are the details for the general case. We won't go though this in class, but please read it carefully.

We now claim that $\boxed{\text{Simpson's Rule}}$ is exact for *any* polynomial of degree 3 or less, and on *any* interval.

Denote by $\boxed{Q_2(f)}$ the approximation of $\int_a^b f(x)dx$ with Simpson's Rule:

$$Q_2(f) = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right).$$

Since the method can be derived by integrating the quadratic that interpolates $f(x)$ at the three points a, $(a+b)/2$, and b, it is clearly exact for all quadratics.

3.1 Simpson's, again

Details

Let $f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$. Then

$$\int_a^b f(x)dx = \int_a^b \left(c_0 + c_1 x + c_2 x^2\right)dx + c_3 \int_a^b x^3 dx =$$
$$\int_a^b \left(c_0 + c_1 x + c_2 x^2\right)dx + c_3 \frac{b^4 - a^4}{4}.$$

Also,

$$Q_2(f) = Q_2(c_0 + c_1 x + c_2 x^2) + Q_2(c_3 x^3) =$$
$$\int_a^b \left(c_0 + c_1 x + c_2 x^2\right)dx$$
$$+ c_3 \left(\frac{b-a}{6}\right)\left(a^3 + 4\left(\frac{a+b}{2}\right)^3 + b^3\right). \quad (1)$$

3.1 Simpson's, again

Details

With a bit of symbolic manipulation we get that

$$\frac{b^4 - a^4}{4} = (\frac{b-a}{6})\left(a^3 + 4(\frac{a+b}{2})^3 + b^3\right),$$

as required.

In the previous section (§3.2 Simpson's Rule), we discussed a naive attempt to derive an upper bound for the error in Simpson's rule leading to

$$\mathcal{E}_2 \leq \frac{(b-a)^4}{196} M_3.$$

This is not wrong – just not sharp. For example it does not give that Simpson's Rule is exact for all cubic polynomials. The sharp result is

Theorem 3.1

$$\mathcal{E}_2 := \left| \int_a^b f(x)dx - Q_2(f) \right| \leq \frac{(b-a)^5}{2880} M_4.$$

For the proof see the text book (Theorem 7.2 of Suli and Mayers). Instead of working through it in class we'll prove a more general version of a consequence it.

3.2 Precision

Definition 3.2 (Precision of a Quadrature Rule)

A quadrature rule has **precision** N if it is exact for all polynomials of degree N or less. That is, the rule $Q(f)$ has precision N if

$$Q(p) = \int_a^b p(x)dx \quad \text{for all } p \in \mathcal{P}_N.$$

Example 3.3

By construction, the $(N+1)$-point Newton-Cotes rule has precision N.

3.2 Precision

Theorem 3.4

If $Q_{2k}(\cdot)$ is a Newton-Cotes quadrature rule on $2k+1$ points, then $Q_{2k}(\cdot)$ has in fact precision $2k+1$.

Proof: Let p be a polynomial of degree $N+1$. We wish to show that $Q_N(p) = \int_a^b p(x)dx$. We can take $a=-1$, $b=1$ because a simple linear transformation can be used to map to an arbitrary interval.

Also, since the quadrature points are equally spaced on $[-1,1]$ we have that $x_i = -x_{n-i}$.

Furthermore (see Exercise 3.1) the quadrature weights are symmetric: $q_i = q_{n-i}$.

3.2 Precision

3.3 Composite Rules

Suppose that we want to estimate $\int_a^b f(x)dx$ and the Trapezium rule is not sufficiently accurate. We could try Simpson's Rule, which should be better. Failing that, we could try a 4-point rule, based on integrating the p_4 interpolant, or a five-point rule, based on integrating p_5.

However, quite apart from the fact that it might be tedious to derive these rules, we know (Runge's example again!) that high-order polynomial interpolation can be very inaccurate.

3.3 Composite Rules

It is better to use a **Composite Rule**. This is analogous to the idea behind piecewise linear interpolation.

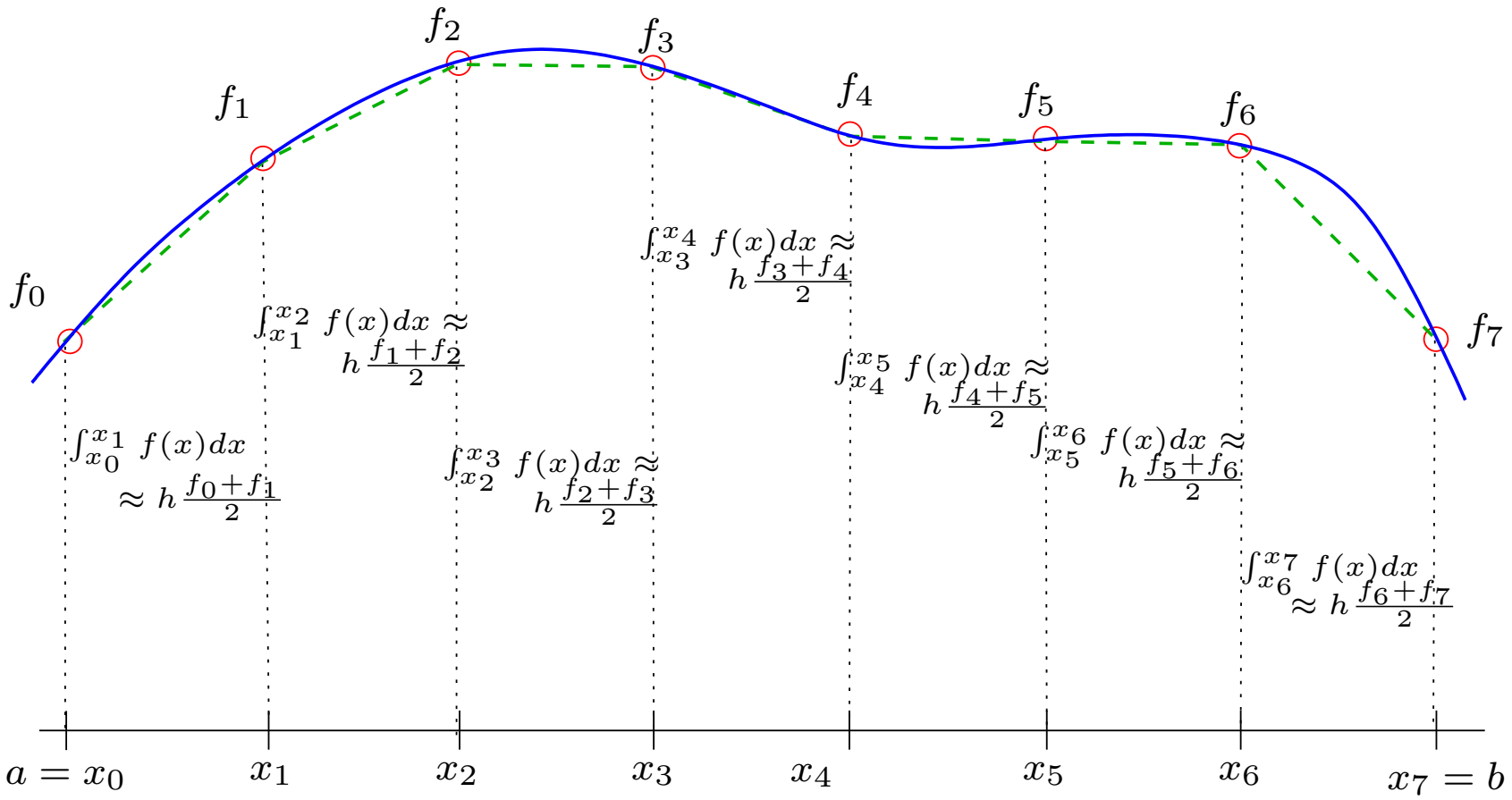
For the **Composite Trapezium Rule** we divide $[a, b]$ into N intervals of size $h = (b - a)/N$. Applying the Trapezium Rule on each interval $[x_{i-1}, x_i]$ we get

$$\int_{x_{i-1}}^{x_i} f(x)dx \approx h\frac{f_{i-1} + f_i}{2}.$$

Summing for the N intervals we get

$$\int_a^b f(x)dx \approx \frac{b-a}{N}\left(\frac{f_0}{2} + f_1 + f_2 + \cdots + f_{N-1} + \frac{f_N}{2}\right). \qquad (2)$$

3.3 Composite Rules



3.4 Exercises

Exercise 3.1

Explain clearly, with an example, why in general it is not true that $Q_n(f) \to \int_a^b f(x)dx$ as $n \to \infty$.

Exercise 3.2

(i) Deduce an error estimate for the Composite Trapezium Rule.

(ii) Taking $N = 10$, give an upper bound for the error in the Composite Trapezium Rule when approximating $\int_1^2 \ln(x)dx$.

(iii) What value of N would you have to take to ensure that the error was less that 10^{-5}?

3.4 Exercises

Exercise 3.3

(i) Deduce the formula for the *composite Simpson's Rule*.

(ii) Derive an error estimate for the *composite Simpson's Rule*.

(iii) What value of N would you have to take to ensure that the error in the estimate of $\int_1^2 \ln(x)dx$ is less that 10^{-6}?

(iv) Denote the $(N+1)$-point Composite Simpson's Rule by $S_N(f) \approx \int_a^b f(x)dx$. Show that, for sufficiently smooth $f(x)$,

$$\lim_{n\to\infty} S_N(f) = \int_a^b f(x)dx.$$

3.4 Exercises

Exercise 3.4

Determine the precision of the following schemes for estimating $\int_0^1 f(x)dx$.

(i) $Q(f) = f(\frac{1}{2})$.

(ii) $Q(f) = \frac{1}{4}f(0) + \frac{3}{4}f(\frac{2}{3})$.

(iii) $Q(f) = \frac{3}{2}f(\frac{1}{3}) - 2f(\frac{1}{2}) + \frac{3}{2}f(\frac{2}{\mathbf{3}})$.

Exercise 3.5

Consider the rule:

$$R(f) = q_0 f(\frac{1}{3}) - f(\frac{1}{2}) + q_2 f(\frac{3}{4})$$

for approximating $\int_0^1 f(x)dx$.

(a) Determine values of q_0 and q_2 that ensure this rule has precision 2.

(b) What is the maximum precision of $R(\cdot)$ with the values of q_1 and q_2 that you have determined?

(c) Why is this not, strictly speaking, a Newton-Cotes rule?